Pharma FP&A & Financial Close Analytics

Roadmap and interview guide · as of 18 October 2026 · Daniel Zhang
路线图与面试指南 · 2026 年 10 月 18 日

One rule for every answer
Say only what you actually did. Numbers in [brackets] are filled in from your real results when each stage is done. If a result differs from the resume draft, change the resume, not the story.
每个回答只说你真正做过的。[方括号] 里的数字等阶段完成后用实际结果填写。如果结果和简历草稿不同，改简历，不改故事。

1. Roadmap at a glance

路线图总览

#	Stage	Status	Done so far	Next
1	Learn the accounting	In progress	Accounting guide with exercises and bilingual glossary	Write your own one-page glossary: docs/glossary.md
2	Design the company	Done	Halvern Pharmaceuticals: 3 products modelled on CMS 2024 Part D data, GTN, cost centers, opening balances	Review and defend every assumption
3	Generate & record 12 months	In progress	Generator pieces 1-4 of 5 (year 2024): 1,047 sales invoices, 1,132 payments, 192 purchase invoices, 78 journal entries	Piece 5: bank lines + Medicaid unwinding scenario; you: ERPNext setup and import
4	Move the data to SQL	Not started	-	Schema script, fact / dimension / versioned budget tables
5	Close controls, tested	Not started	Error plan designed: 4 types x 10, held-out test	7 control queries, close workbook, seeded test, score
6	Variance & GTN reporting	Not started	-	Price / volume split, GTN bridge, 3-page dashboard, timing log
7	Package & publish	In progress	GitHub repo, folder layout, README, Chinese pipeline	Merge to main, screenshots, results, resume links

2. The 60-second pitch

一分钟项目介绍

"This is a self-directed project on how a pharma finance team closes its books and explains its results. Real ledgers are confidential, so I built a simulated drug company whose products, prices and volume mix come from real 2024 Medicare Part D data I'd already cleaned. I generated twelve months of transactions, recorded them in ERPNext, moved them to SQL, and built two things on top: close controls that catch errors, tested on a held-out ledger with planted errors, and budget-versus-actual and gross-to-net reporting in Power BI. The lesson I took away: balanced books aren't the same as correct books."

“这是我自主做的项目，研究制药公司财务部怎么结账、怎么解释业绩。真实账本是保密的，所以我搭建了一家模拟药企，产品、价格和销量结构来自我清洗好的真实 2024 年 Medicare Part D 数据。我生成了十二个月的交易，在 ERPNext 记账，导入 SQL，然后做了两件事：能发现错误的月结控制（在植入错误的留出账本上测试），以及 Power BI 中的预算对比和毛到净报告。我的收获：账平不等于账对。”

3. Stage by stage: what to say

逐阶段：面试怎么说

Stage 1: Learn the accounting [In progress]

Learn the five concepts every control relies on: general ledger, AR/AP, accruals, bank reconciliation and gross-to-net.

What to say

"I started by learning the accounting the controls depend on: double entry, subledgers versus control accounts, accruals, bank reconciliation and gross-to-net. In pharma, gross-to-net matters most, because a large share of gross sales goes to rebates and chargebacks that are only estimates when they're booked."

“我先学习了控制所依赖的会计基础：复式记账、明细账与控制科目、应计、银行对账和毛到净。在制药行业毛到净最重要，因为很大一部分毛销售额会变成返利和价格补差，而记账时这些都只是估计数。”

Likely questions

Q: Walk me through the entry for a rebate. **A:** Month-end: Dr Rebates (contra-revenue) / Cr Rebate accrual (liability). When paid: Dr Rebate accrual / Cr Bank. Revenue is reduced in the month of the sale, not the month of payment.
Evidence to show: Your glossary; the accounting guide exercises.

Stage 2: Design the company [Done]

Halvern Pharmaceuticals, fictional. Glyvanta, Respiqor and Onkalyn are modelled on Jardiance, Trelegy Ellipta and Xtandi using real 2024 Medicare Part D price per claim, with volumes scaled by 1/1000. About $20M gross sales a year.

What to say

"I designed the company from real data. In an earlier project I cleaned CMS Medicare Part D data, 288.7 billion dollars of 2024 drug spending. I picked three drugs from different therapeutic areas, kept their real price per claim and scaled volumes down by a thousand. A script reproduces every number. The gross-to-net rates are my assumptions, and I can explain each one: pharmacy drugs in competitive classes pay large rebates, while the oncology drug pays small rebates but more chargebacks."

“公司是用真实数据设计的。我在之前的项目里清洗了 CMS Medicare Part D 数据（2024 年共 2,887 亿美元药品支出），从不同治疗领域选了三个药，保留真实的单次处方价格，销量缩小一千倍。每个数字都能用脚本重现。毛到净扣减率是我的假设，每一个我都能解释。”

Likely questions

Q: Why fictional products instead of the real drugs? **A:** Real companies' internal books are confidential. Real prices and volume mix keep it realistic; fictional names make clear I'm not claiming anyone's actual financials.
Q: What does your data not cover? **A:** Part D covers pharmacy-dispensed drugs only, not hospital-administered ones (Part B), so the chargeback rate is an assumption, not data. All CMS figures are gross, before rebates.
Evidence to show: docs/company-design.md, data/design_inputs.py.

Stage 3: Generate & record 12 months [In progress]

A Python generator creates about 400 transactions a month (sales invoices, payments, purchases, journal entries, bank lines) with a fixed random seed, then they're imported into ERPNext, which writes the general ledger.

What to say

> *"The transactions are synthetic. I generated them with a Python script with a fixed seed, so anyone can reproduce them, and built in realistic behaviour: weekly wholesaler order patterns, Net 30 and Net 45 payment timing, monthly accruals. Then I loaded them into ERPNext, so the general ledger was created by a real accounting system that enforces balanced entries and linked documents, not by my script."*
>
> “交易是模拟的。我用固定随机种子的 Python 脚本生成，任何人都能重现，并加入了真实的业务规律：批发商每周下单、Net 30 和 Net 45 的付款时间、每月应计。然后导入 ERPNext，总账是由真实的会计系统生成的，而不是我的脚本。”

Likely questions

Q: Why go through ERPNext instead of writing the ledger directly? **A:** ERPNext enforces accounting rules: balanced entries, invoices linked to payments, closed periods. It also gives the ledger the same structure a real company's ERP export would have.

Q: How many transactions, and how do you know? **A:** Say the actual final count from your row counts by month, not a target number.

Evidence to show: data/generate_transactions.py, ERPNext screenshots, row counts by month.

Stage 4: Move the data to SQL [Not started]

Export the ERPNext general ledger and documents into fact and dimension tables, plus a versioned budget table (v1 original, v2 reforecast). Keep a schema script so the database can be rebuilt from scratch.

What to say

> *"I modelled the data as a star schema: general-ledger and budget facts joined to accounts, cost centers, products and periods. The budget table is versioned, so I can compare actuals against the original budget or the reforecast. The whole database rebuilds from one script."*
>
> “我把数据建成星型模型：总账和预算事实表，连接科目、成本中心、产品和期间维度表。预算表有版本号，可以和原始预算或滚动预测对比。整个数据库一个脚本就能重建。”

Likely questions

Q: Why a versioned budget? **A:** Companies reforecast mid-year. Without a version column you can't say which plan a variance is measured against.

Evidence to show: sql/01_schema.sql, table row counts.

Stage 5: Close & reconciliation controls, tested [Not started]

Seven SQL controls (trial balance, subledger-to-GL, bank match, duplicates, cutoff, classification, unapplied payments) feed an exception register in an Excel Power Query close workbook. Tested on a held-out ledger with 40 seeded errors: 10 duplicates, 10 wrong periods, 10 misclassifications, 10 unmatched payments.

What to say

"A balanced trial balance doesn't prove the books are right, so I built seven controls, each a SQL query whose output rows are exceptions. To test them honestly I wrote the rules on a development copy, then ran them once, unchanged, on a held-out copy with 40 planted errors and an answer key. They found [X] of 40 with [Y] false positives. The ones they missed were [type], because [reason]."

"试算平衡不代表账对，所以我建了七个控制，每个是一个 SQL 查询，输出的行就是异常。为了诚实地测试，我在开发副本上写规则，然后在植入 40 个错误的留出副本上只运行一次、不改规则。找到了 [X] 个，误报 [Y] 个。漏掉的是 [类型]，原因是 [原因]。"

Likely questions

Q: What does 'held-out' mean? **A:** I never looked at the test copy while writing rules, and didn't change any rule after running it. Like practising on old exams, then sitting a new one.
Q: Why use synthetic data to test controls? **A:** You need to know the right answer. With planted errors and an answer key I can measure recall and false positives; with real books you never know every error.
Evidence to show: sql/02_controls.sql, tests/answer_key.csv, score output, exception register.

Stage 6: Variance & gross-to-net reporting [Not started]

Budget vs actual by month, product and cost center; price / volume split; GTN deductions reconciled to the ledger; three Power BI pages; a management report; manual vs automated preparation timed with a stopwatch.

What to say

"The dashboard explains why results differ from plan. Revenue variance is split into volume and price, and the two always add back to the total. The gross-to-net bridge walks from gross sales through each deduction to net revenue, and every bar ties to its ledger account. I timed the monthly package both ways on the same data: [M] minutes manually, [N] automated."

"看板解释为什么结果和计划不同。收入差异拆成量差和价差，两者加起来正好等于总差异。毛到净桥接图从毛销售额经过每一项扣减到净收入，每一项都和总账科目核对。同一数据上我分别计时：手工 [M] 分钟，自动化 [N] 分钟。"

Likely questions

Q: Walk me through a price / volume split. **A:** Volume = (actual units - budget units) x budget price. Price = (actual price - budget price) x actual units. Example: budget 1,000 at $100, actual 1,100 at $95: volume +10,000, price -5,500, total +4,500.
Q: What did the 90 / 20 minutes consist of? **A:** Describe your actual steps from the timing log: export, lookups, rebuild pivots versus refresh, review, write commentary.
Evidence to show: Dashboard pages, management report, report/timing_log.csv.

Stage 7: Package & publish [In progress]

GitHub repo with data, SQL, tests, workbook, dashboard and report; screenshots; links on the resume.

What to say

"Everything is on GitHub: the generator, the SQL, the answer key and scoring query, the timing log. Anyone can rebuild the data and rerun the test and get the same numbers."

"所有内容都在 GitHub 上：生成脚本、SQL、答案表和评分查询、计时记录。任何人都能重建数据、重跑测试，得到同样的数字。"

Likely questions

Q: What would you do differently? **A:** Prepare an honest answer once done, e.g. start the controls earlier, or test on a second seeded ledger.
Evidence to show: github.com/linbei2323-web/reph-1

4. Tough questions

难题

Question	How to answer
Is this a real company?	No. It's a simulated company in a self-directed project. Prices and volume mix come from real CMS data; the transactions are synthetic. Say so plainly: interviewers respect it.
Where does the data come from?	Two layers: real 2024 CMS Medicare Part D data for prices and mix (from my market analysis project), and script-generated transactions recorded in ERPNext.
What was hardest?	Pick a real moment from your own work, e.g. a control that flagged too many false positives and how you fixed it on the development copy.
How does this connect to your market analysis project?	That project stops at gross spending, before rebates. This one covers what comes after gross: gross-to-net, the close and variance analysis. Together they cover both halves of pharma finance.

5. Numbers to fill in from real results only

只能用实际结果填写的数字

Claim	Status today	Source of the final number
Transactions recorded	Generator so far: 2,449 documents (bank lines still to come)	Actual final row count
Errors found / false positives	Not run yet	Your score output
Report time manual / automated	Not measured yet	Your stopwatch log
Project dates	Started Oct 2026	The months you actually worked on it

Takeaway
Balanced accounts do not necessarily mean correct accounts. Reliable reporting requires checks on classification, timing, source documents and reconciliation, not just matching totals.
账平不等于账对：可靠的报告需要检查分类、时间、原始单据和对账，而不只是总数相等。